

LSEG Quantitative Analytics

Database Schema for Northfield Risk Factors

Version 1.0.3



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1 Introduction

In This Document

This document provides schema information about the **Northfield Risk Factors** data tables in Quantitative Analytics.

NOTE

For information about the other Quantitative Analytics tables, refer to the other Database Schema documents posted on the [Quantitative Analytics](#) product page on [MyAccount](#).

The documentation for each table includes these main elements:

- **Vendor Database or License Name** – This element provides the vendor database or license name. Vendor databases generally correspond to vendor licensing options.
- **Table Name: Table Title** – This element provides the table name and title, followed by a description of the table contents.
- **Update Cycle** – This element indicates the frequency of updates from the data vendor. (For specific information about when this update is posted for your use, refer to your update logs. The update log schema description is available in the Database Schema for Quantitative Analytics Core Tables document.)
- **Adjusted** – This element indicates whether the data is stored adjusted, unadjusted, or if adjustment is not applicable (N/A).
- **Indexes, Index Fields** – This element lists the table's index(es) (for example, pkey_CSGCAD) and the index fields. The primary index contains a "pkey_" or "PK_" prefix. Clustered indexes are noted.
- **Field, Type, Nullable, Definition** – This element describes the table's fields, including field names, data types, and whether each field is nullable. The data types in Snowflake have equivalents available. For example, an *int* in SQL Server is a *number* in Snowflake.

Intended Readership

This document is intended for Quantitative Analytics clients licensed to use **Northfield Risk Factors** data through LSEG. Note that the tables described in this document may be covered by several licenses; your access to the data in these tables is dependent on the licenses you have. Readers should have basic knowledge of Microsoft SQL Server 2016.

Table Name Conventions

In Quantitative Analytics, database tables are grouped by name. Generally, the first few letters of the table name refer to the data vendor or licensing group. For example, all tables that begin with **NF*** are **Northfield Risk Factors** data tables.

Pervasive and Oracle

This document is specific to Microsoft SQL Server 2016. If you are accessing the data directly through the Pervasive database, you may encounter slight differences in index names and data types. Please note that references to Pervasive and Actian are synonymous.

If you are accessing the data through Oracle, note that field names that correspond to Oracle reserved words are appended with an underscore. For example, field names "Number", "Synonym", and "UID" become "Number_", "Synonym_", and "UID_" in Oracle.

Zero Values and Nulls

Null values are usually recorded as "NULL". Occasionally, a table may contain a value of "-99999" or "-9999"; in these cases, there is no recorded data for that period. (It is null.) If the table contains a "0" value, the data was reported as having a "0" value.

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 - **RIC Change Events** inform you of planned changes to RICs. RIC is the market level identifier key unique to LSEG used to identify securities.
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2. Northfield Risk Factors Tables

NFCmCode: Common Integer Coded Values

This table contains integer coded values.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFCmCode (Clustered)		Type_, Code	
Field	Type	Nullable	Description
Type_	smallint	N	Type_ is a numeric code for a type of data.
Code	int	N	Code is a coded value for an element of Type_. When Type_ = 0, Code is the numeric code for each Type_ other than 0 and Desc_ is a description of each Type_.
Desc_	varchar(127)	Y	Desc_ is a plain-language description of the element defined by Type_ and Code.

NFCmDesc: Common String Coded Values

This table contains string coded values. This table can be joined to the [NFCmCode](#) table using the following comparisons in the query:

- NFCmCode.Type = NFCmDesc.Type
- NFCmCode.Desc_ = NFCmDesc.Code

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFCmDesc (Clustered)		Type_, Code	
Field	Type	Nullable	Description
Type_	smallint	N	Type_ is a numeric code for a type of data.
Code	varchar(4)	N	Code is a coded value for an element of Type_. When Type_ = 0, Code is the numeric code for each Type_ other than 0 and Desc_ is a description of each Type_.
Desc_	varchar(127)	Y	Desc_ is a plain-language description of the element defined by Type_ and Code.

NFCmFactors: Common Risk Factors Information

This table contains risk factors information.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFCmFactors (Clustered)		FactorCode	

Field	Type	Nullable	Description
FactorCode	smallint	N	FactorCode is a unique integer code that identifies a factor. This integer code is defined in TRDW and is not received from the vendor.
FactorID	varchar(30)	N	FactorID is the risk factor identifier.
Name_	varchar(25)	N	Name_ is the risk factor name.
LongName	varchar(100)	N	LongName is the risk factor long name.
ModelType	smallint	N	ModelType specifies the type of model to which the risk factor belongs. There are five types of risk models in Northfield Risk Factors: – Domestic – Global – Regional – Single Country – Everything Everywhere NFCmFactors.ModelType cross-references NFCmCode.Code where Type_ = 7. Use the following query: <pre>select C.Desc_ from NFCmFactors F join NFCmCode C on F.ModelType=C.Code and C.Type_=7 where F.FactorCode=73</pre>

NFCmIndustry: Common Industry Information

This table contains industry information.

Update Cycle: Monthly

Adjusted: No

Indexes	Index Fields
pkey_NFCmIndustry (Clustered)	IndustryCode

Field	Type	Nullable	Description
IndustryCode	smallint	N	IndustryCode is a unique integer code that identifies an industry. This integer code is defined in TRDW and is not received from the vendor.
SectorType	smallint	Y	SectorType distinguishes between three different types of industry data available in Northfield Risk Models. – 1 = Northfield US Fundamental Sector (domestic industry data): FND/APT Industry mnemonics are integer values. – 2 = Northfield US Real Estate Sector (US real estate industry data): REIT model industry mnemonics are integer values. – 3 = Northfield Global Sector (global industry data): In global industry data, industry Mnemonics are in Sx-Cy format. For example, in s3-c23, s3 means sector code 3, while c23 means country code 23. Both country codes and sector codes are defined in NFCmCode and NFCmDesc. NFCmIndustry.SectorType cross-references with NFCmCode where Type_ = 5.
Mnemonic	varchar(10)	Y	Mnemonic is the industry identifier. Each sector type has a separate naming convention for the industry mnemonic.
Name_	varchar(100)	Y	Name_ is the industry name.
SectorCode	smallint	Y	SectorCode is the sector code to which the industry belongs. NFCmIndustry.SectorCode cross-references with NFCmCode.Code where Type_ = 10.

NFCmInfo: Common Entity Information

This table contains a single entry for each entity encountered, including the latest value of entity identifiers.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFCmInfo (Clustered)		Code	
Field	Type	Nullable	Description
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TDW and is not received from the vendor.
Name_	varchar(150)	Y	Name_ is the security name.
NFNIS	varchar(12)	Y	NFNIS is the Northfield unique security identifier. The vendor files contain Northfield unique identifiers for equity data only. Most of the securities in the EE model are without unique identifiers.
NFTicker	varchar(12)	Y	NFTicker can be either a security identifier or a Northfield custom identifier. The value of the NFIdType field determines what NFTicker represents: – NFIdType = 1 for a security identifier – NFIdType = 2 for a Northfield custom identifier
Cusip	varchar(9)	Y	Cusip is the security's CUSIP identifier.
Sedol	varchar(7)	Y	Sedol is the security's SEDOL identifier.
Isin	varchar(12)	Y	Isin is the security's ISIN identifier.
UpdateDate	datetime	Y	UpdateDate is the most recent date the record was updated.
NFIdType	smallint	Y	NFIdType is used to distinguish between custom IDs and security ticker IDs in the NFTicker field.
AssetType	smallint	Y	AssetType is not populated from the vendor file.

NFCmModel: Common Risk Model Information

This table contains risk model information.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFCmModel (Clustered)		ModelCode	
Field	Type	Nullable	Description
ModelCode	smallint	N	ModelCode is a unique integer code that identifies a risk model.
Mnemonic	varchar(8)	N	Mnemonic is the risk model mnemonic.
Name_	varchar(25)	N	Name_ is the risk model name.
ModelType	smallint	N	ModelType specifies the type of model to which the risk factor belongs. There are three types of risk models in Northfield Risk Factors: – 1 = Domestic – 2 = Global – 3= Regional – 4 = Single Country – 5 = Everything Everywhere NFCmModel.ModelType cross-references with NFCmCode .Code where Type_ = 7.

NFCmSecIdent: Common Security Identifier Change History

This table contains security identifier change history information.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFCmSecIdent (Clustered)		Ident, IdentType, StartDate	
Field	Type	Nullable	Description
Ident	varchar(12)	N	Ident contains the change history of the NFTicker, CUSIP, SEDOL, ISIN, and NFId of a security.
IdentType	smallint	N	IdentType describes what type of security identifier is present in the Ident field. IdentType can be: – 1 = CUSIP – 2 = SEDOL – 3 = FI_Index – 4 = Ticker – 5 = ISIN – 6 = NFId
StartDate	datetime	N	StartDate is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
EndDate	datetime	Y	EndDate is populated when a security is changed or discontinued. The date is in the format YYYYMMDD.
Code	int	Y	Code is the unique value assigned to a security in the NFCmInfo table.

NFModCor: Model Factor Correlations

This table contains correlations between two factors by model.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFModCor (Clustered)		Date_, ModelCode, CurrCode, FactorCode1, FactorCode2	
Field	Type	Nullable	Description
Date_	datetime	N	Date_ is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is an integer code that identifies the Northfield Risk Model. NFModCor.ModelCode cross-references with NFCmCode .Code where Type_ = 6.
CurrCode	smallint	N	CurrCode is an integer code that identifies the currency. NFModCor.CurrCode cross-references with NFCmCode .Code where Type_ = 8.
FactorCode1	smallint	N	FactorCode1 is the integer code assigned to factor1; it is the first factor with respect to which correlation is stored. This is a foreign key from the NFCmFactors table.
FactorCode2	smallint	N	FactorCode2 is the integer code assigned to factor2; it is the second factor with respect to which correlation is stored. This is a foreign key from the NFCmFactors table.
Corr	float	N	Corr is the correlation value between FactorCode1 and FactorCode2.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModCor.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModFactor: Model Factor Order

This table contains the factor order for each model.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFModFactor (Clustered)		StartDate, ModelCode, FactorCode	
Field	Type	Nullable	Description
StartDate	datetime	N	StartDate is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is a unique integer value assigned to Northfield Risk Models.
FactorCode	smallint	N	FactorCode is a unique integer code that identifies a factor.
EndDate	datetime	Y	EndDate is the date on which the data in the file is invalid. The date is in the format YYYYMMDD.
FactorOrder	int	Y	FactorOrder is the order of factors for each model in the Exposure and Correlation File (not the field number).
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModFactor.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModSec: Model Security Information

This table contains security information for each model.

Update Cycle: Monthly (Daily for Short Horizon Models)

Adjusted: No

Indexes		Index Fields	
pkey_NFModSec (Clustered)		Date_, ModelCode, Code	
Field	Type	Nullable	Description
Date_	datetime	N	Date_ is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is a unique integer value assigned to Northfield Risk Models.
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TDW and is not received from the vendor. This value is also populated from the NFCmSecIdent table.
SecPrice	float	Y	SecPrice is the security price.
MthResdRisk	float	Y	MthResdRisk is the stock-specific risk.
NumShare	float	Y	NumShare is shares outstanding (in millions).
TotalRisk	float	Y	TotalRisk is the total risk of the security.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModSec.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModSecCntryCurr: Model Country and Currency Information

This table contains country and currency information for each security in a model.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFModSecCntryCurr (Clustered)		StartDate, ModelCode, Code	

Field	Type	Nullable	Description
StartDate	datetime	N	StartDate is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is an interger code that identifies the Northfield Risk Model. NFModSecCntryCurr.ModelCode cross-references with NFCmCode .Code where Type_ = 6.
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TRDW and is not received from the vendor. This value is also populated from the NFCmSecIdent table.
EndDate	datetime	Y	EndDate is populated when a security is changed or discontinued. The date is in the format YYYYMMDD.
ISOCntry	char(2)	Y	ISOCntry is the country with which the model and security are associated.
CurrCode	smallint	Y	CurrCode is the currency with which the model and security are associated. NFModSecCntryCurr.CurrCode cross-references with NFCmCode .Code where Type_ = 8.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModSecCntryCurr.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModSecExposure: Model Security Exposure

This table contains the security exposure to a factor by model.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFModSecExposure (Clustered)		Date_, ModelCode, Code, FactorCode	

Field	Type	Nullable	Description
Date_	datetime	N	Date_ is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is an integer code that identifies the Northfield Risk Model. NFModSecExposure.ModelCode cross-references with NFCmCode .Code where Type_ = 6.
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TDW and is not received from the vendor. This value is also populated from the NFCmSecIdent table.
FactorCode	smallint	N	FactorCode is an integer code assigned to a factor. FactorCode is a foreign key from the NFCmFactors table.
FacExposure	float	N	FacExposure is the factor exposure value (factor coefficient), for the factor represented by FactorCode.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModSecExposure.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModSecExposureEE: Model Security Exposure – Everything Everywhere

This table contains the security exposure to a factor for the Everything Everywhere (EE) model.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFModSecExposureEE (Clustered)		Date_, ModelCode, Code, FactorCode	
Field	Type	Nullable	Description
Date_	datetime	N	Date_ is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is an integer code that identifies the Northfield Risk Model.
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TRDW and is not received from the vendor. This value is also populated from the NFCmSecIdent table.
FactorCode	smallint	N	FactorCode is an integer code assigned to a factor. FactorCode is a foreign key from the NFCmFactors table.
FacExposure	float	N	FacExposure is the factor exposure value (factor coefficient), for the factor represented by FactorCode.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModSecExposure.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModSecIndustry: Model Sector and Industry Information

This table contains sector and industry information for securities by model.

Update Cycle: Monthly

Adjusted: No

Indexes		Index Fields	
pkey_NFModSecIndustry (Clustered)		StartDate, ModelCode, Code, SectorType	
Field	Type	Nullable	Description
StartDate	datetime	N	StartDate is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is an integer code that identifies the Northfield Risk Model. NFModSecIndustry.ModelCode cross-references with NFCmCode .Code where Type_ = 6.
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TDW and is not received from the vendor. This value is also populated from the NFCmSecIdent table.
SectorType	smallint	N	SectorType distinguishes between three different types of industry data available in Northfield Risk Models. 1 = Northfield US Fundamental Sector (domestic industry data): FND/APT Industry mnemonics are integer values. 2 = Northfield US Real Estate Sector (US real estate industry data): REIT model industry mnemonics are integer values. 3 = Northfield Global Sector (global industry data): In global industry data, industry Mnemonics are in Sx-Cy format. For example, in s3-c23, s3 means sector code 3, while c23 means country code 23. Both country codes and sector codes are defined in NFCmCode and NFCmDesc. NFCmIndustry.SectorType cross-references with NFCmCode where Type_ = 5.

Field	Type	Nullable	Description
EndDate	datetime	Y	EndDate is populated when a security is changed or discontinued. The date is in the format YYYYMMDD.
IndustryCode	smallint	N	IndustryCode is the industry code with which the security of the model is associated. IndustryCode is a foreign key from NFCmIndustry table.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFMModSecIndustry.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModSector: Model Sector Codes

This table contains sector codes by model.

Update Cycle: Monthly

Adjusted: No

Indexes	Index Fields
pkey_NFModSector (Clustered)	ModelCode, SectorType

Field	Type	Nullable	Description
ModelCode	smallint	N	ModelCode is a unique integer value assigned to Northfield Risk Models.
SectorType	smallint	N	SectorType distinguishes between three different types of industry data available in Northfield Risk Models. – 1 = Northfield US Fundamental Sector (domestic industry data): FND/APT Industry mnemonics are integer values. – 2 = Northfield US Real Estate Sector (US real estate industry data): REIT model industry mnemonics are integer values. – 3 = Northfield Global Sector (global industry data): In global industry data, industry Mnemonics are in Sx-Cy format. For example, in s3-c23, s3 means sector code 3, while c23 means country code 23. Both country codes and sector codes are defined in NFCmCode and NFCmDesc. NFCmIndustry.SectorType cross-references with NFCmCode where Type_ = 5.

NFModSecUniverse: Model Security Historical Timeframes

This table contains historical timeframes of securities by model.

Update Cycle: Monthly **Adjusted:** No

Indexes	Index Fields
pkey_NFModSecUniverse (Clustered)	StartDate, ModelCode, Code

Field	Type	Nullable	Description
StartDate	datetime	N	StartDate is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is an integer code that identifies the Northfield Risk Model. NFModSecUniverse.ModelCode cross-references with NFCmCode .Code where Type_ = 6.
Code	int	N	Code is a unique integer code that identifies a security. This integer code is defined in TDW and is not received from the vendor. This value is also populated from the NFCmSecIdent table.
EndDate	datetime	Y	EndDate is populated when a security is discontinued. The date is in the format YYYYMMDD.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModSecUniverse.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

NFModVar: Model Factor Variance

This table contains the variance of factors by model.

Update Cycle: Monthly (Daily for Short Horizon Models)

Adjusted: No

Indexes	Index Fields
pkey_NFModVar (Clustered)	Date_, ModelCode, CurrCode, FactorCode

Field	Type	Nullable	Description
Date_	datetime	N	Date_ is the date on which the data in the file is valid. The date is in the format YYYYMMDD.
ModelCode	smallint	N	ModelCode is a unique integer value assigned to Northfield Risk Models.
CurrCode	smallint	N	CurrCode is a unique integer code that identifies a currency. NFModVar.CurrCode cross-references with NFCmCode .Code where Type_ = 8.
FactorCode	smallint	N	FactorCode is an integer code assigned to a factor. FactorCode is a foreign key from the NFCmFactors table.
AnnVar	float	N	AnnVar is the annual variance value of a factor.
LicFlag	int	Y	LicFlag is a bit value that represents the license for this table. NFModVar.LicFlag cross-references with NFCmCode .Code where Type_ = 12.

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